

SCIENCE - WEEK 1

The Senses, and What Makes Something Alive

Two topics this week: how your body senses the world, and how to tell living things from non-living things

Name: _____

Week starting: _____

Level: foundation (PreK-K, paired). These topics lead into later Biology: the nervous system, and the characteristics of living things.

For the Grown-Up

Why two topics this week. These are foundation-level topics she has likely met informally before, so we're pairing them to move a little faster and free up time later for the harder Grade 6-8 material.

What she will learn. By the end of the week she should be able to: name the five senses and their organs and explain how nerves and the brain are involved; give the 7 characteristics that scientists use to decide if something is alive; and sort things around her into living, non-living (was never alive), and dead (once alive).

Time. About four to five sessions of 35-45 minutes.

How to use "See it first!" boxes. Before each new section, look up the search phrase together (SafeSearch on) so the words that follow land on something she has already pictured.

Materials. A mirror; a rubber band; a small potted plant or houseplant; a rock or pebble; a toy that once had batteries (or any toy); access to a window or the outdoors for a nature walk; pencil and colored pencils.

Suggested week

Day 1	The senses: sight and hearing, with experiments.
Day 2	Smell, taste, touch, and the brain as the control centre.
Day 3	The 7 characteristics of living things.
Day 4	Living, non-living, and dead: sorting and a nature walk.
Day 5	Research and report, review quiz, check with the answer key.

Topic 1: How Your Body Senses the World

SEE IT FIRST!

Search: **five senses and the nervous system diagram**

Look at how the sense organs connect to the brain. What carries the message between them?

Every second, your body is collecting information: light bouncing off this page, the hum of a fridge, the smell of dinner cooking. Your body collects this information using five **sense organs**: eyes (sight), ears (hearing), nose (smell), tongue (taste), and skin (touch).

Each sense organ contains **receptors**, tiny structures that detect a specific kind of information, such as light, sound waves, chemicals, or pressure. When a receptor is triggered, it sends an electrical signal along a **nerve**. Nerves are part of your **nervous system**, a network that carries messages around your body at incredible speed, up to about 120 metres per second.

All these signals travel to your **brain**, which interprets them and decides how to respond. Sometimes the response is a thought ("that smells like popcorn"), and sometimes it's an automatic reaction called a **reflex**, like blinking or pulling your hand off something hot before you've even consciously felt the pain.

DID YOU KNOW?

A reflex reaction can happen before your brain has even finished processing the pain. Some reflex signals only travel as far as your spinal cord and back, which is faster than sending a message all the way to the brain.

ACTIVITY: Reflex and Reaction

1. Sit facing a partner. Have them hold their hand out, palm up, just under yours, without touching.
2. Without warning, they try to catch your hand as you snatch it away. Swap roles.
3. Now test a real reflex: gently tap just below the kneecap with the side of your hand while the leg dangles freely. Does the leg twitch forward on its own?

Record what you noticed about how fast your body reacted without you 'deciding' first.

Eye experiment: In a dim room, look at your eyes in a mirror and note the pupil size. Switch on a bright light and watch again. Explain in your own words why the pupil changes size.

The Senses in Depth

SEE IT FIRST!

Search: **how does the human ear detect sound waves**

Search: **taste buds and smell receptors working together diagram**

Hearing. Sound is caused by vibrations travelling through the air as waves. The outer ear funnels these waves to the **eardrum**, a thin membrane that vibrates. Three tiny bones pass the vibration to a fluid-filled, snail-shaped structure called the **cochlea**, where receptors turn it into nerve signals.

Smell and taste. These two senses work as a team, which is called **chemoreception**, meaning they detect chemicals. Smell receptors high in the nose detect airborne chemical particles. Taste buds on the tongue detect five basic tastes: sweet, sour, salty, bitter, and umami (savory). Most of what we call "flavour" is actually smell, which is why food seems bland with a blocked nose.

Touch. Skin is the body's largest organ and contains several different kinds of receptors: for light pressure, deep pressure, temperature, and pain. Fingertips and lips have the densest concentration of receptors, which is why they are the most sensitive.

THINK LIKE A SCIENTIST

Why might it be useful, from a survival point of view, that pain reactions can happen faster than conscious thought?

People who are deaf or blind often report that their other senses feel sharper. Can you think of a reason why that might be, even though the sense organ itself hasn't changed?

Topic 2: Living vs Non-Living

SEE IT FIRST!

Search: **7 characteristics of living things for kids**

Look at the examples. Which of the seven can you already name before reading on?

At first, deciding what is "alive" seems easy: an animal is alive, a rock is not. But some things are trickier, like a seed that looks dry and still, or a candle flame that moves and grows. Scientists use seven characteristics to decide whether something is truly a living thing. Something needs to do **all seven**, not just one or two, to count as alive.

Movement	Living things move all or part of themselves, even plants, which slowly turn toward light.
Respiration	Living things release energy from food inside their cells (this is not the same as just breathing).
Sensitivity	Living things detect and respond to changes around them, called stimuli.
Growth	Living things get bigger and develop over their lifetime.
Reproduction	Living things can produce offspring, more of their own kind.
Excretion	Living things remove waste products made inside their bodies.
Nutrition	Living things take in and use food or make their own, for energy and materials.

Memory trick: the first letters spell **MRS GREN** — Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition.

DID YOU KNOW?

A candle flame moves, grows, gives off waste gases, and needs 'fuel' to keep going, so it can seem alive by a couple of clues. But it cannot reproduce, cannot sense and respond to its surroundings, and isn't made of cells, so it fails the full MRS GREN test. It is non-living.

Living, Non-Living, or Dead?

There are actually **three** categories to sort things into, and people often mix up the last two:

- **Living:** currently doing all seven MRS GREN characteristics right now, e.g. a tree, a bird, a person
- **Dead:** was once alive and could do the seven characteristics, but no longer can, e.g. a fallen leaf, a dead insect, a cooked carrot
- **Non-living (never alive):** has never been alive and never could do the seven characteristics, e.g. a rock, water, a toy car, a cloud

ACTIVITY: Sort It Out

Go on a walk around your home, garden, or a park. Find at least 10 things. Sort each into Living, Dead, or Non-living, and for the tricky ones, write which MRS GREN test they fail.

Thing I found	Living / Dead / Non-living	Why (for tricky ones)

Tricky Cases

SEE IT FIRST!

Search: **is a virus alive or non-living debate**

Search: **seed dormancy how a dry seed is still alive**

Case 1: A dry seed. A seed sitting in a packet looks completely still. It isn't visibly moving, growing, or doing anything. But inside, it is alive in a resting state called **dormancy**. Its life processes are happening extremely slowly, and with water, warmth, and time, it will sprout.

Case 2: A virus. Viruses are one of the strangest cases in biology. They can reproduce, but only by invading a living cell and hijacking it; on their own, they can't grow, respire, or respond to anything. Because they fail most of the MRS GREN tests on their own, most scientists classify viruses as non-living, though it's genuinely debated.

THINK LIKE A SCIENTIST

Is a chicken egg alive? Think about which of the seven characteristics apply to what's growing inside it.

Is fire alive? Go through MRS GREN and decide which ones it passes and which it fails.

Research and Report

Choose ONE question to research, then teach your family what you learned, like presenting to a class.

- Are viruses alive or not? Find two reasons scientists give for each side of the debate.
- How do bats 'see' in the dark using their ears instead of their eyes? What is this called?
- How does a Venus flytrap plant sense and respond to touch, even though it has no nervous system?

My chosen question: _____

Facts I found (at least 3):

My summary (write 4-5 sentences):

[] I presented this to my family and answered a question about it.

My Week 1 Review

1. Fill in the blank.

The five sense organs are the eyes, ears, nose, tongue, and _____.

Nerves carry messages to the _____, which interprets them.

An automatic, very fast response like blinking is called a _____.

The seven characteristics of living things are remembered with the letters _____.

A leaf that has fallen off a tree and dried up is an example of something that is _____.

2. Short answers.

1) Name the seven characteristics of living things, in full (not just the letters).

2) Explain why a candle flame is not considered a living thing.

3) Explain why a dry, dormant seed is still considered alive.

4) Describe the path a signal takes from your finger touching something hot, to you feeling it.

Answer Key and Notes (Grown-Ups)

Fill in the blank: skin; brain; reflex; MRS GREN; dead (once alive, no longer able to do MRS GREN).

Short answers: 1) Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition. 2) It moves, grows, and gives off waste gases, but cannot reproduce, cannot sense and respond to its environment, and is not made of cells. 3) Its life processes are still happening, just extremely slowly; with water, warmth and time it will resume visible growth. 4) Skin receptors detect the heat and send an electrical signal along a nerve, often triggering a fast reflex at the spinal cord that pulls the hand away before the pain signal has even finished reaching the brain.

Watch for. A common mix-up is calling anything that was once alive 'non-living' — encourage the three-way sort (living / dead / non-living) rather than a simple two-way one. Also watch for 'respiration' being confused with just breathing; it's the release of energy from food inside cells.

Discussion starters. The virus question has no single right answer and is a good one to let her sit with. For the chicken egg: a fertilised egg with a developing embryo is alive; the yolk and white alone are not.

Next week (Week 2): Plants Around Us.